Name : AKASH SAMPATH

Topic : Docker Commands

User and Navigation Commands :

Command	**Meaning**
`sudo su -`	Switch to root user admin
`cd /home`	Change directory to /home folder
`ls`	List all files and folders
`touch akash.txt`	Create an empty file
`exit`	Exit / Log out / Root/User

Docker Image Commands :

Command	**Meaning**
`docker pull nginx`	Download nginx image from Docker Hub
`docker images`	Show all downloaded images on your system
`docker rmi nginx`	Remove / Delete the nginx image
`docker build -t akashdockeraws/akashdocker1008:ubuntuimage .`	Create custom image from a Docker file
`docker login`	Log in to Docker Hub account
`docker push akashdockeraws/akashdocker1008:ubuntuimage`	Upload your custom image to Docker Hub

Container Management Commands :

Commands	Meaning
`docker run -dt –name akash1 nginx`	Create and start a container name akash1 from nginx image
`docker ps`	Show only running containers
`docker ps -a`	Show all containers + Stopped ones
`docker stop akash1`	Stop a running container
`docker rm akash1`	Remove or delete stopped container
`docker exec -it akash1 bash`	Open a terminal inside running container

File Transfer Commands :

Commands	Meaning
`docker cp /home/akash.txt container_id:/home`	Copy file from computer into a container

Port Mapping Commands :

Commands	Meaning
docker run -dt -p 13:80 --name akash3 nginx	Map port 13 on your computer to port 80 in container access nginx at localhost:13

Docker Volumes Commands :

Commands	Meaning
`docker volume create akashvolume`	Create a name volume for storing data
`docker volume ls`	List of all volumes
`docker run -dt --mount source=akashvolume,destination=/home --name akash1 nginx`	Run container with volume attached

Docker Networks Commands :

Commands	Meaning
`docker network create --driver=bridge --subnet=192.168.0.0/24 akashbridge`	Create a custom network for containers to communicate
`docker run --network=akashbridge --name akash1 nginx`	Run container on specific network
`docker inspect akash1`	Show detailed info about containers

Docker File : Commands :

Commands	Meaning
`FROM ubuntu:latest`	Base image to start from
`RUN apt update`	Run command during image build
`RUN apt install apache2 -y`	Install Apache web Server
`RUN mkdir /dockerfile`	Create a folder
`RUN touch /dockerfile/akash.txt`	Create a file
`EXPOSE 77 / : wq!`	Document that container listens on port 77 / Save and exit in vi editor